# The `Polyhedral` submodule of `dmplex`: user documentation and test examples

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**Abstract**

`dmplex.Polyhedral` carries the finite-volume method of the Fortran code POLYEDER over to Julia: cell complexes of general polyhedra, the exact integration of monomials over cells and faces, an average-preserving polynomial reconstruction, upwind transport, face gradients from interaction regions, the shallow-water equations on one layer of cells over the sphere (KernelAbstractions, MPI, GPU), and cut cells over orography. This document describes how the pieces are used and shows the results of the test examples. The derivations are in the companion paper draft `docs/polyhedral/polyhedral.tex`.

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## 1 Overview

The data flow is the same for every application: a `PolyComplex` (from an `ExtrudedPlex` or cut by a surface), its `PolyGeometry`, operators assembled once as sparse matrices (reconstruction, face values, gradients), and a `tendency!(dy, y, model)` for the explicit time steppers of `dmplex` (`SSPRK3Cache`, `ssprk3_step!`). All functions are in the module `dmplex.Polyhedral`; the exported ones are listed in `Polyhedral.jl`, the others are reached as `dmplex.Polyhedral.name` (below `PH.name`).

| file (`src/Polyhedral/`) | contents |
|---|---|
| `Complex.jl` | `PolyComplex`, `extruded_complex`, `face_cells`, `vertex_neighbours` |
| `Integration.jl` | `MonomialBasis`, `cell_moments`, `face_moments`, `face_quadrature` |
| `Geometry.jl` | `PolyGeometry`: volumes, centroids, areas, normals |
| `Reconstruction.jl` | `PolyReconstruction`, `face_value_matrices`, `point_value_matrices` |
| `Advection.jl` | `PolyAdvection`, `stream_function_flux`, `vector_potential_flux` |
| `InteractionRegion.jl` | `interaction_operators`, `cell_vectors`, `linear_vector_reconstruction` |
| `ShallowWater.jl` | `PolyShallowWater`: kernels, MPI distribution, time step |
| `CutCell.jl` | `cut_complex`, `merge_cells`, `merge_small_cells` |
| `Euler.jl` | `PolyEuler`: compressible Euler equations on extruded cells, IMEX |

Table 1: Files of the submodule.

## 2 Cell complexes

A `PolyComplex` stores cells, faces, edges and vertices, numbered from 1 in each dimension, with compressed cones and incidence numbers $\pm 1$:

- the faces of cell $c$ are `cell_faces[cell_offsets[c]:cell_offsets[c+1]-1]`, with `cell_signs` $+1$ when the face normal points out of the cell;
- the edges of a face, likewise in `face_offsets`, `face_edges`, `face_signs`, the orientation of the face (and so its normal) given by the edge loop;
- `edge_vertices` ($2\times$edges, $\partial e = v_2 - v_1$) and `coordinates`.

`PH.cell_cone(cx, c)` and `PH.face_cone(cx, f)` return the cone and its signs, `PH.face_loop(cx, f)` the vertices of a face in order. `face_cells(cx)` gives the two cells of every face ($2\times$faces): the one the normal points out of, then the one it points into, 0 on the boundary. Faces need not be planar: a non-planar face is integrated over the fan of triangles from its vertex mean (`planar_pieces`).

The complexes used so far are prisms over a 2D mesh:

```
using dmplex
plex = create_hexagon_mesh(3)    # or create_spherical_mesh(3),
    create_cubed_sphere_mesh(8)
cx   = extruded_complex(ExtrudedPlex(plex, 1), [0.0, 1000.0]; radius=6.37122e6)
geo  = PolyGeometry(cx)                     # volume, cell_centroid, area, normal,
     face_centroid
```

On the sphere vertex $v$ of level $k$ is at $(R+z_k)\,v$, on a plane at $v + z_k\,\hat{z}$; a plane mesh comes from `plex_from_polygons(polygons, points)`. Figure 1 shows one layer over the three spherical meshes of the tests.

## 3 Moments and geometry

`MonomialBasis(p)` lists the exponents $\alpha$ of the monomials $y^\alpha$ of degree $\le p$ in three variables. `cell_moments(cx, c, B; center, scale, rotation)` returns the integrals $\int_C y^\alpha\, dV$ in the local coordinates $y = Q(\mathbf{x}-\mathbf{x}_0)/h$, exactly, by homogeneous integration down the cone hierarchy (cell $\to$ faces $\to$ edges $\to$ vertices); `face_moments` does the same for a face. Without `center` and `scale` a frame of the cell (or face) itself is used, which keeps the recursion well conditioned. `PolyGeometry` takes volumes and centroids from the moments of degree $\le 1$, the unit normal of a face from its area vector. `face_quadrature(cx, f)` gives points and area vectors of a rule of degree 5 on a face (the seven-point rule on every triangle of the fan).

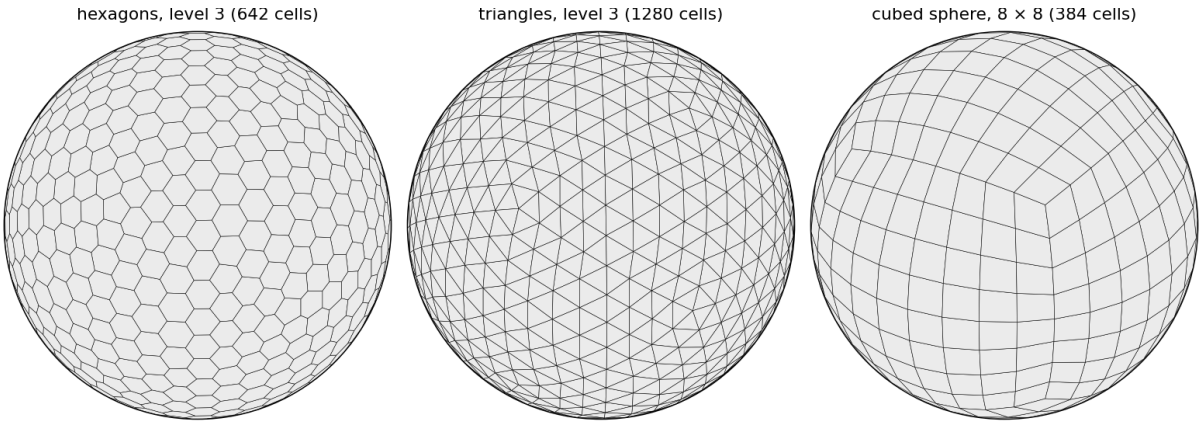



Figure 1: One layer of prisms over the sphere: the bottom faces of the hexagonal, the icosahedral (triangles) and the cubed-sphere mesh.

The test set "cell complexes and the integration of monomials" checks the moments against exact values on the unit cube, a triangular prism and a non-convex L-shaped prism, in shifted and scaled frames, the divergence theorem for every monomial, the closure $\sum_F s_{CF}|F|\mathbf{n}_F = 0$ and the volume of a spherical shell over icosahedra, and the closure of hexagonal prisms with non-planar faces.

# 4 Reconstruction and transport

## 4.1 Reconstruction

```
rec = PolyReconstruction(cx, geo, degree; basis=:full, rings=nothing,
                         oversampling=1.5, boundary_oversampling=oversampling,
                         up=normalize, rtol=1e-8)
Q1, Q2 = face_value_matrices(cx, rec)     # faces x cells: face averages of side 1
     and 2
reconstruct(rec, c, u, x)                  # value of the polynomial of cell c at
    x
```

For every cell the polynomial of degree `degree` that has the cell average and fits the averages of the stencil in the least-squares sense. The stencil grows by vertex rings until the problem has full rank and `oversampling` times as many cells as monomials; `boundary_oversampling` is used for the cells with a boundary face, whose stencils are one-sided (the cut cells at the terrain need 2.5 for degree 2). `basis=:tangential` drops the monomials in the direction `up(x_C)`: the reconstruction in the tangent plane for one layer over the sphere. The coefficients are linear in the averages, so face values and point values are sparse matrices (`face_value_matrices`, `point_value_matrices`).

## 4.2 Transport

```
U = stream_function_flux(cx, ψ; G=r -> r^3/3)   # volume fluxes through the faces
m = PolyAdvection(cx, U; degree=2, basis=:tangential)
dt = 0.4 * upwind_time_step(m)
cache = SSPRK3Cache(q)
ssprk3_step!(q, dt, PH.tendency!, cache, m)
```

`PolyAdvection(cx, flux)` transports cell averages with the upwind face averages of the reconstruction; with a velocity function instead of the fluxes it integrates $(\mathbf{u}\cdot\mathbf{n})\,q$ at the quadrature

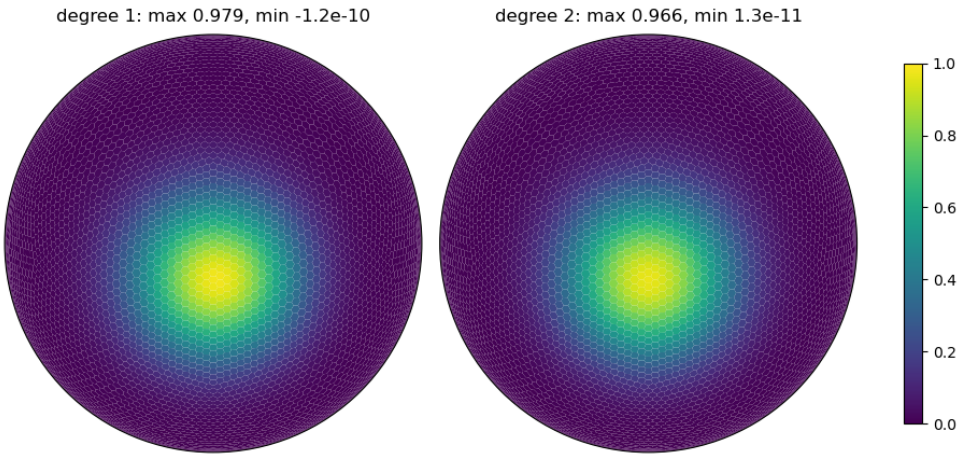



Figure 2: Solid-body rotation of a Gaussian hill on hexagons, level 5, after one revolution (587 steps of SSP-RK3), degree 1 and 2 of the reconstruction.

points of the faces, each point upwinded, which gives the order of the reconstruction (degree 3: order 4 in the test). Divergence-free fluxes come from a potential: `stream_function_flux(cx, ψ; G, height, project)` for $\mathbf{u} = g(h)\nabla\psi \times \nabla h$ on the sphere (solid-body rotation: $\psi = \Omega \cdot \hat{r}$, $G(r) = r^3/3$; with the default $G(r) = r$ the rotation on a shell $1 \le r \le 1.01$ is about 1 % too slow), and `vector_potential_flux(cx, A)` for $\mathbf{u} = \nabla \times A$ in general, the circulation of $A$ around every face. In both cases the outward fluxes of every cell sum to zero to round-off.

Figure 2 shows a Gaussian hill $q = \exp(-|\hat{r} - \hat{x}|^2/0.16)$ after one revolution about the axis $(1, 0, 1)/\sqrt{2}$ on hexagons of level 5 (10242 cells). The relative $L^2$ errors are 0.120, 0.033 (degree 1) and 0.120, 0.022 (degree 2) on levels 4 and 5; mass is conserved to $10^{-13}$.

## 5 Interaction regions and cell vectors

```
G, T = interaction_operators(cx, geo)   # faces x cells: normal gradient, face
    value
vc  = cell_vectors(cx, geo, un)         # Perot: vector at the cells from normal
    components
R   = cell_vector_matrices(cx, geo)     # the same as three sparse matrices
R   = linear_vector_reconstruction(cx, geo; own_weight=20)   # linear fit, second
    order
un  = face_normal_components(cx, geo, vc)
```

Around every vertex the cells and faces meeting there form an interaction region (POLYEDER `TGrad`); a gradient per cell, continuous normal components across the faces of the region, gives the normal gradient `G` and the face value `T` as sparse matrices, exact for linear fields at faces away from the boundary. `cell_vectors` is Perot's reconstruction $v_C = |C|^{-1} \sum_F s_{CF} |F|\, u_F\, (\mathbf{x}_F - \mathbf{x}_C)$, exact for constant fields only; `linear_vector_reconstruction` fits a linear field to the normal components of the cell and its neighbours (own faces weighted by `own_weight`) and is second order on triangles as well.

## 6 Shallow water on one layer

```
m = PolyShallowWater(cx; gravity=9.80616, omega=7.292e-5, bathymetry=nothing,
                     degree=2, cell_vector=:auto, comm=nothing, backend=CPU())
```

```
y = initial_state(m, velocity, height)       # local [h; U], U = h u.n at the
    faces
dt = shallow_water_time_step(m, y; cfl=0.4)
cache = SSPRK3Cache(y)
ssprk3_step!(y, dt, tendency!, cache, m)
state = gather_state(m, y); total_mass(m, y); M = cell_momentum!(m, y)
```

Heights are cell averages, momenta normal components at the faces. The pressure gradient comes from `interaction_operators`, advection and Coriolis act on the cell momentum (Perot, or the linear fit: `cell_vector=:linear`, chosen by `:auto` on triangular prisms), whose Cartesian components are transported with the upwind face values of the reconstruction and projected back to the faces. All terms are KernelAbstractions kernels over padded tables; with `backend = CUDABackend()` they run on the GPU. With an MPI communicator `comm` the cells are partitioned by area, ghosts are the stencil columns of the owned rows, and the result is bitwise equal to the serial run (test `test/mpi_poly.jl`, 1 and 3 ranks).

**Galewsky et al. (2004).** `examples/polyhedral/galewsky_poly.jl`, configured by environment variables (`MESH=icosahedral|hexagonal|cubed LEVEL DAYS CV PERTURB GPU OUT`), runs the barotropic jet; on Levante one A100 per case (`poly_galewsky.sbatch`, set `SLURM_CPUS_PER_TASK=16`). Six days at level 7 take 1–2 minutes on the GPU, the setup of the operators on the host 10–14 minutes. Figure 3 shows the relative vorticity at day 6. The perturbed jet rolls up as expected on all three meshes; the unperturbed jet should stay zonal, but breaks up into waves of the grid's wavenumber 5 by day 6: the open problem of the one-layer model. The pressure gradient of the interaction regions is first order in the maximum norm near the irregular vertices (second order in the mean), the current suspect.

# 7 Cut cells

```
level(x) = x[3] - zs(x[1], x[2])                 # > 0 in the fluid
cut, parent, terrain = cut_complex(cx, level)   # parent cell, terrain faces
g0 = PolyGeometry(cx)
merged, group = merge_small_cells(cut, g0.volume[parent]; fraction=0.5)
```

`cut_complex` follows POLYEDER's `CutGrid`: vertices with `level < 0` are removed, edges between the fluid and the ground are cut at the root of `level` (Illinois method), every cut face is closed by the edge between its two cut points, every cut cell by a terrain face through these edges (non-planar in general, oriented outward). A face crossed twice by the surface, or a vertex exactly on it, is an error. `merge_small_cells` unites every cell smaller than `fraction` times its reference volume with the neighbour across its largest face, repeatedly; `group` maps the cut cells to the merged ones, and a merged cell is again a polyhedron of the complex, so moments, reconstruction and transport apply unchanged.

The test (Figure 4) cuts jittered triangular prisms over $[-1,1]\times[0,\frac{1}{4}]\times[0,1]$ by $z_s = 0.03 + 0.25\,e^{-(x/0.4)^2}\cos^2(\pi x/0.5)$. Unmerged, the smallest cell has $2.5\cdot10^{-11}$ of its prism (inside the domain, not on the wall shown); merged at 0.5 the steady transport $q = e^{2\psi}$ by $\mathbf{u} = \nabla\times(\psi\hat{y})$, $\psi = 0$ on the terrain and the top, runs with 26 instead of 74638 steps on $16\times8$ and converges with orders 1.76 and 1.91 ($16\times8 \to 32\times16 \to 64\times32$).

## 7.1 Flow over a hill on a Cartesian cut-cell grid

`examples/polyhedral/mountain.jl` solves the compressible Euler equations on a Cartesian grid of hexahedra (one cell across in $y$) cut by the hill and merged at 0.5. Prognostic are the cell averages of $\rho$ and $\rho\theta$ and the normal momentum $M_F = (\rho\mathbf{u})\cdot\mathbf{n}_F$ at the faces (a C grid; zero

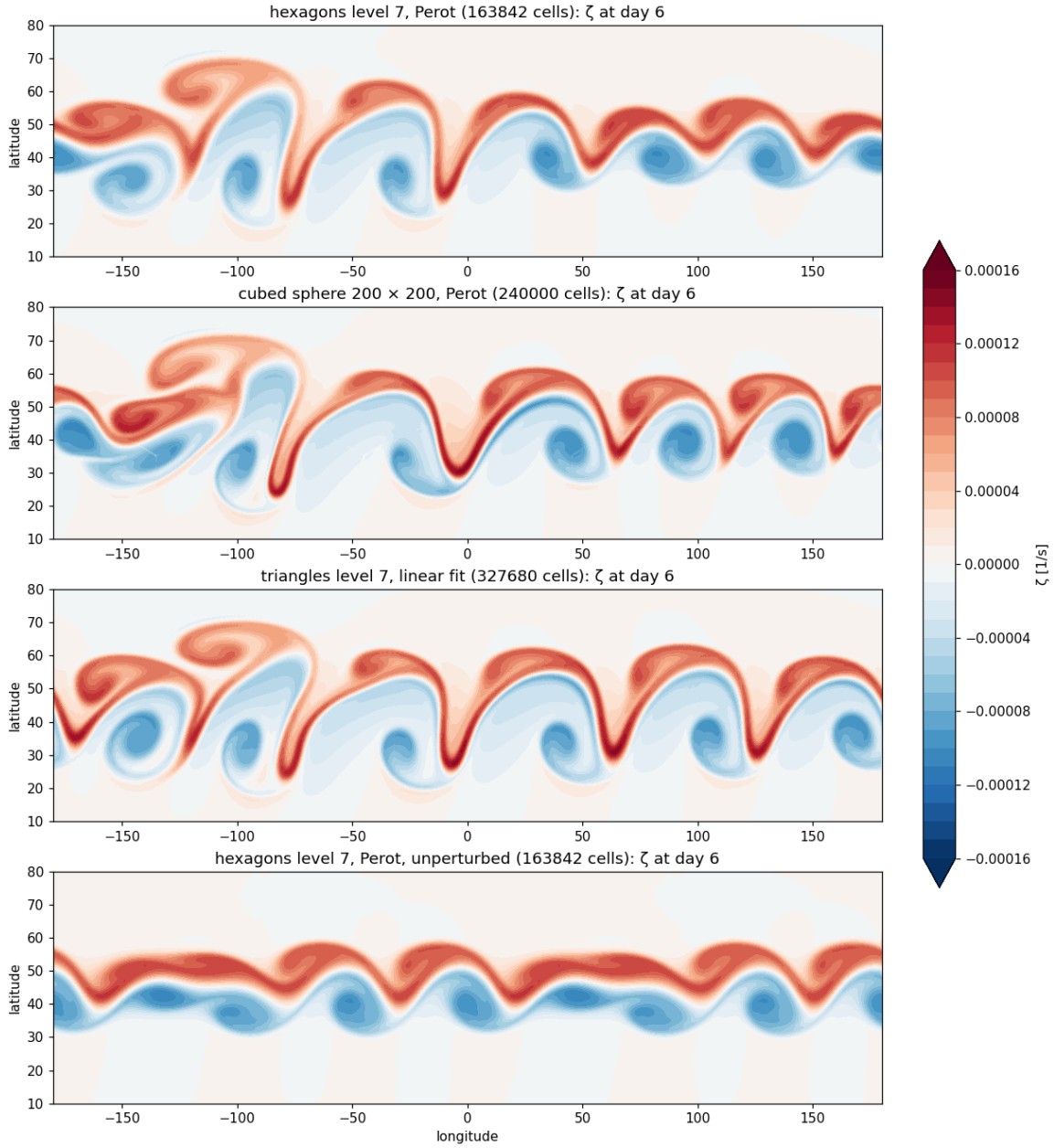



Figure 3: Galewsky test, relative vorticity at day 6: perturbed jet on hexagons (Perot), the cubed sphere ($200 \times 200$ per face, Perot) and triangles (linear fit), and the unperturbed jet on hexagons.

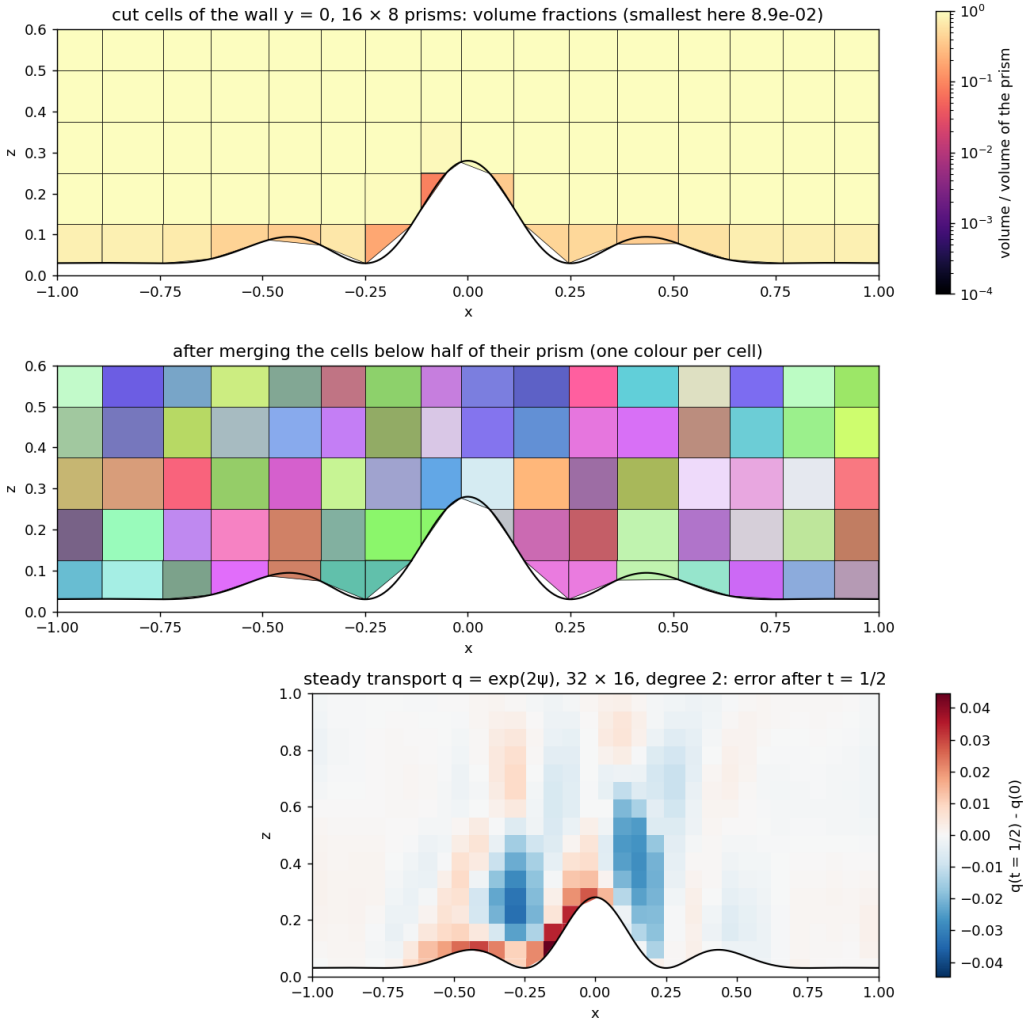



Figure 4: Cut cells on the wall $y = 0$ of the test: volume fractions of the cut cells ($16 \times 8$), the cells after merging, and the error of the steady transport ($32 \times 16$, degree 2, $t = 1/2$).

at the terrain, the top and the side walls, prescribed at the lateral boundaries):

$$\partial_t \rho_C = -|C|^{-1} \textstyle\sum_F s_{CF} |F| M_F, \qquad \partial_t (\rho\theta)_C = -|C|^{-1} \textstyle\sum_F s_{CF} |F| M_F \, \theta_F^{\uparrow},$$

$$\partial_t M_F = \mathbf{n}_F \cdot (w_1 A_{C_1} + w_2 A_{C_2}) - \frac{p'_{C_2} - p'_{C_1}}{\mathbf{n}_F \cdot (\mathbf{x}_{C_2} - \mathbf{x}_{C_1})} - g\, n_{F,z} \, (w_1 \rho'_{C_1} + w_2 \rho'_{C_2}) - \tau_F \, (M_F - M_F^0),$$

with $p = p_{00}(R_d \rho\theta / p_{00})^\gamma$, the perturbations $p'$, $\rho'$ against the discrete initial state at rest (which is therefore exactly balanced), $A_C = -|C|^{-1} \sum_F s_{CF} |F| M_F \, \mathbf{u}_F^{\uparrow}$ the momentum advection of the cell velocity $\mathbf{u}_C = m_C / \rho_C$ (Perot's $m_C$), $\uparrow$ the upwind face values of the degree-2 reconstruction, $w_{1,2}$ the linear interpolation weights along the normal, and Rayleigh damping $\tau$ above $z_D$ and near the lateral boundaries. The initial flow is the discrete curl of $A = -\psi \hat{y}$ with $\psi = U_0 \, [\, R(z) - R(z_s) \max(0, (z_d - z)/(z_d - z_s)) \,]$, $R(z) = \int_0^z \rho_0$, $z_d = 5$ km: divergence free, tangential to the terrain, uniform aloft. SSP-RK3 with the acoustic time step.

The test of Gallus and Klemp (2000): Agnesi hill $h(x) = h_0/(1 + (x/a)^2)$, $h_0 = 400$ m, $a = 1$ km, $U_0 = 10$ m/s, $N = 0.01$ s$^{-1}$ ($Na/U_0 = 1$, nonhydrostatic, $Nh_0/U_0 = 0.4$), $\theta_s = 300$ K, $\Delta x = 250$ m, $\Delta z = 125$ m. Figure 5 compares $w$ and $\theta'$ with the linear solution (Fourier integral with radiation condition and the amplitude growth $\sqrt{\rho_0(0)/\rho_0(z)}$). The domain is $50 \times 20$ km ($200 \times 160$ Cartesian cells, 31970 after cutting and merging), with damping above 12 km and beyond $|x| = 17$ km; $\Delta t = 0.152$ s, 71276 steps for 3 h, 68 minutes on 8 threads of a Mac mini.

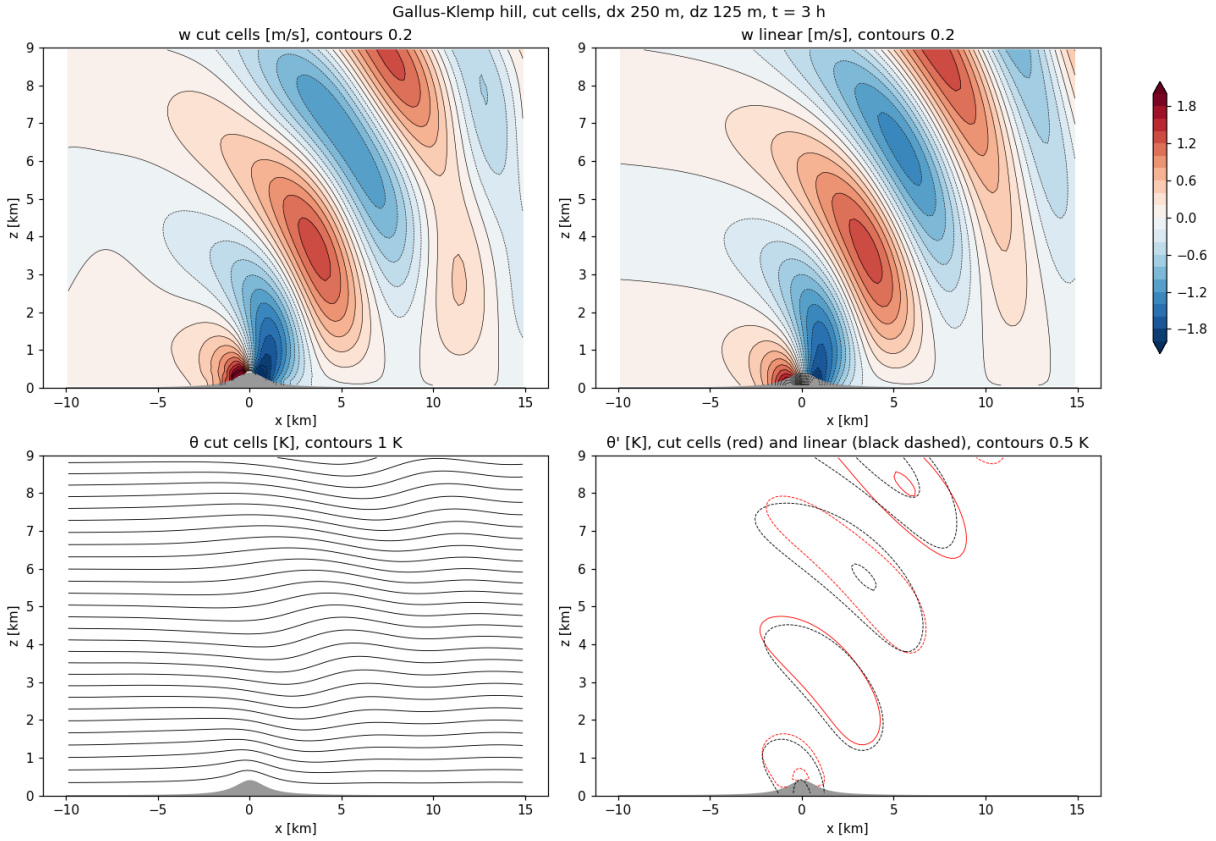



Figure 5: Flow over the Gallus–Klemp hill on the Cartesian cut-cell grid: $w$ of the model and of the linear solution, $\theta$, and $\theta'$ of the model (red) and the linear solution (black dashed).

At $t = 3$ h the maximum of $|w|$ is 2.36 m/s (linear 2.40), at $z = 2$ km $w$ lies between $-1.36$ and 0.89 m/s (linear $-1.33$, 0.92), and the phase lines of $w$ and $\theta'$ follow the linear solution; the isentropes pass smoothly over the hill, without noise from the cut cells. The mass changes by less than $2 \cdot 10^{-5}$, only through the relaxation in the damping layers.

# 8 The Euler equations on extruded cells

```
v  = VerticalGrid(30, 30000.0)
cx = extruded_complex(ExtrudedPlex(create_hexagon_mesh(5), 30), v.z_half; radius
    =6.37122e6)
m  = PolyEuler(cx, v.z_half; radius=6.37122e6, damping_height=22000.0,
    damping_rate=0.05,
                degree=1, projection=:adjoint)
y  = initial_state(m, (λ, φ, z) -> baroclinic_wave(λ, φ, z))
ys = zero_state(m)
step!(m, y, 300.0, ys)
surface_pressure(m, y); lowest_temperature(m, y); vertical_velocity(m, y);
    total_mass(m, y)
```

`PolyEuler` is the mountain scheme of Section 7.1 on the prisms of an `extruded_complex`, over the sphere or the plane: $\rho$ and $\rho\theta$ at the cells, the normal momentum $M_F$ at the faces, $\theta$ and the velocity upwind from the reconstruction of degree `degree`, the pressure gradient between the two cells along the normal and the buoyancy of the deviations $p'$, $\rho'$ from an isothermal state at rest. It has the interface of the other dycores (`initial_state(m, profile)`, `zero_state`, `step!`, `total_mass`, `surface_pressure`, `lowest_temperature`, `vertical_velocity`), so the

test cases of `Nonhydrostatic` run with it through `examples/dycores.jl` (`baroclinic_wave.jl poly`, `thermal_wind.jl poly`). Four choices were needed for the baroclinic wave:

1. *Heights from the levels.* The faces of the polyhedra are planar, so the bottom of a cell sags below the sphere by $d^2/2a$ for cells of size $d$ (90 km at hexagon 2, 1 km at hexagon 5); the radius of a point is no height. A cell has the height of the middle of its layer, a face between layers that of its level, and the distances between layers are those of the levels, so all columns are alike: a horizontally uniform state at rest stays at rest exactly (test set). Gravity is constant and acts across the faces between layers, the shallow atmosphere of the test cases.

2. *Traditional Coriolis force* $-f\hat{r} \times m_C$, $f = 2\Omega \cdot \hat{r}$: the full $-2\Omega \times m$ forces the steady wave vertically by $2\Omega u \cos\varphi \approx 3 \cdot 10^{-3}$ m/s$^2$, which the shallow test case does not balance.

3. *The adjoint projection of the cell forces.* The forces of the cells (advection of the cell momentum, Coriolis) come to the faces by the adjoint of Perot's reconstruction $R$, $\big[(\mathbf{x}_F - \mathbf{x}_1) \cdot A_1 + (\mathbf{x}_2 - \mathbf{x}_F) \cdot A_2\big]/\mathbf{n} \cdot (\mathbf{x}_2 - \mathbf{x}_1)$, not interpolated along the normal. The face momenta have a null space of $R$, about one mode per cell (three lateral faces against two horizontal components; the spectrum of $P =$ interpolation $\circ\, R$ is real in $[0, 1]$ with 477 zeros on 1440 lateral faces of hexagon 2); its divergence-free part feels no pressure, and an interpolated force drives it linearly: the steady wave drifted by 3.5 m/s per day at hexagon 4 and blew up by day 5. The adjoint keeps the forces orthogonal to it (the pressure gradient is the negative adjoint of the divergence), and the Coriolis force is energy neutral, $\sum_C |C|\, m_C \cdot (-f\hat{r} \times m_C) = 0$ (interpolated it lets noise grow at the poles). It is first order where the line between the centroids is not normal to the face.

4. *The vertical acoustics linearized every step.* ARS(2,3,2) (Ascher, Ruuth and Spiteri 1997) treats the fluxes through the faces between layers, their pressure gradient, buoyancy and damping implicitly, linear in $y - y_{\text{ref}}$ with the sound speed $\gamma p/\rho\theta$ and $\theta$ at the faces of the state at the beginning of the step. Linearized once about the initial state, the explicit remainder of the vertical sound waves has the speed $c\sqrt{\gamma\, \Delta\theta/\theta}$, 40 m/s once $\theta$ changed by 1 %: the step was limited to 20 s at 1 km layers. Per step the system is solved column by column: $\rho'$, $\rho\theta'$ eliminated, a tridiagonal system for the upward momentum (Thomas, threaded over the columns), exact to $5 \cdot 10^{-13}$ against the dense solve. The scheme is stable for horizontal Courant numbers up to 1 (the HEVI amplification of the 2D acoustics).

Figure 6 compares the baroclinic wave with `mpas` at the same resolution (hexagon 5, 30 layers, dt 300 s). With the upwind momentum transport the minimum surface pressure is 994.4, 988.3, 976.1, 961.2 and 955.6 hPa at days 6 to 10 (mpas 992.6, 983.5, 968.4, 946.9, 921.8), and by day 10 poly has five equal waves around the globe instead of one wave packet, the fivefold symmetry of the icosahedral mesh, while mpas, icon and the kite model agree on one packet near 120° E. The imprinting is not stronger than in mpas: without the perturbation the largest zonal standard deviation of $p_s$ over 5° bands at day 10 is 12.7 hPa (mpas 14.6 hPa), but the wave of poly grows too slowly to dominate it. The upwinding of the momentum is the main damping: transported centred (`momentum_upwind=0`) the minimum is 992.6, 984.4, 964.6, 947.9 and 938.1 hPa with a hyperviscosity $\nu_4 = 3 \cdot 10^{15}$ m$^4$/s (`hyperviscosity`, the default of `dycores.jl`, $\propto \Delta x^3$), 934.4 hPa at day 10 without it; the hyperviscosity of the cell velocity does not reduce the imprinting (13.4 hPa with $\nu_4 = 10^{16}$). Mass is conserved to $10^{-15}$; 241 ms per step on 8 threads of a Mac mini after a setup of 9 minutes (mostly the reconstruction). With `degree=2` the run blows up within the first day (open). The thermal wind (`thermal_wind.jl poly 4 5 300`) keeps $\max|w|$ at $1.6 \cdot 10^{-3}$ m/s for 5 days, the temperature of the lowest layer changes by 0.19 K and the surface pressure by 0.13 hPa.

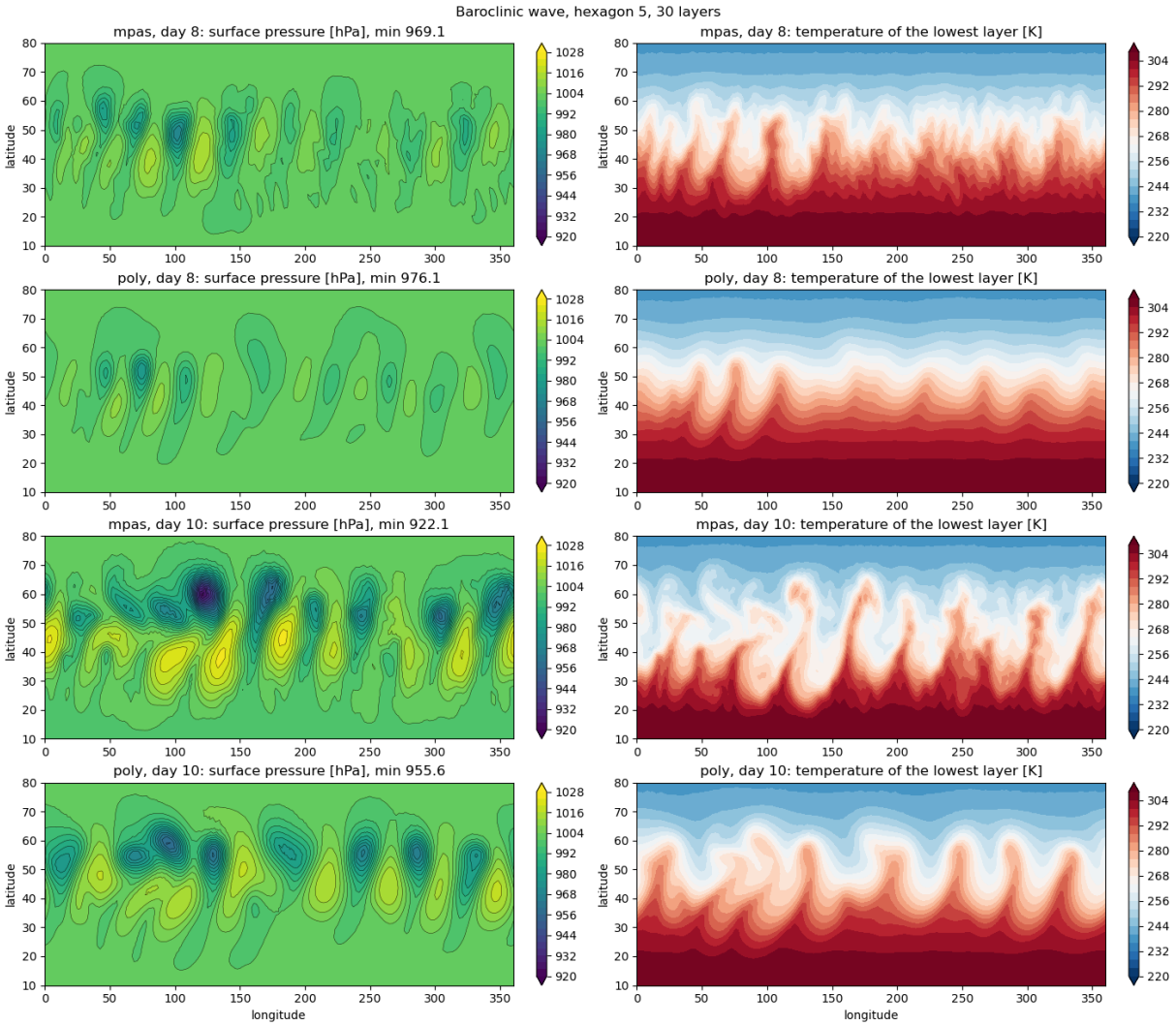



Figure 6: Baroclinic wave (Ullrich et al. 2014) on hexagon 5 with 30 layers, surface pressure and temperature of the lowest layer at days 8 and 10: MPAS-Gassmann (`mpas`) and `PolyEuler` (`poly`), from `baroclinic_wave.jl -out` and `examples/plot_surface.py`.

# 9 Local mass matrices on polygons

The two-point gradient of `PolyEuler` is consistent only where the line between the centroids is normal to the face (on hexagon 4 the angle is 0.45° on average and at most 2°). This section collects the attempts at a consistent gradient that keeps the energy pairing $G = -B^T$ and a local (explicit) mass matrix.

**Multipoint gradient** The gradient of the interaction regions (`interaction_operators`, Section 5) in place of the two-point gradient of `PolyEuler` blows up within the first day on hexagon 4, with and without the perturbation of the baroclinic wave ($\rho\theta < 0$). It is not the adjoint of the divergence, so there is no energy bound, and the regions are three-dimensional: on the lateral faces they pick up the large hydrostatic vertical gradient of $p'$ from the cells above and below.

**Corner unknowns** The prototype `examples/polyhedral/corner/corner_sw.jl` (one layer, shallow water) has the normal momentum on the two halves of every lateral face, one per corner (vertex column). At the corner $(C, v)$ the two half faces of $C$ give the full vector $\mathbf{u}_{Cv} = N_{Cv}^{-1} U$, the inner product is the corner quadrature $\sum_{C,v} \omega_{Cv}\, \mathbf{u}_{Cv} \cdot \mathbf{v}_{Cv}$ with the subcell areas $\omega_{Cv}$, so

$M = C^T W C$ is block diagonal per vertex (the vertex rule of Wheeler and Yotov, the lumped $\mathrm{BDM}_1 \times P_0$ of Egger and Radu on triangles); the gradient is $M^{-1} B^T$, the Coriolis force $-f\hat{\mathbf{r}} \times \mathbf{u}_{Cv}$ at the corners does no work. The linear system conserves energy (largest real part of the eigenvalues $4 \cdot 10^{-19}$). Williamson 2 after 5 days, relative $L^2$ error of $h$, with `PolyShallowWater` ("C grid") for comparison:

| | hexagons 3 | hexagons 4 | icosahedral 3 | icosahedral 4 |
|---|---|---|---|---|
| corner unknowns | 0.145 | 0.157 | $9.3 \cdot 10^{-3}$ | $2.1 \cdot 10^{-3}$ |
| C grid | $2.7 \cdot 10^{-3}$ | $6.7 \cdot 10^{-4}$ | $2.1 \cdot 10^{-3}$ | $5.7 \cdot 10^{-4}$ |

On triangles the scheme converges at second order (the super-convergence of the pressure of the lumped $\mathrm{BDM}_1$), four times less accurate than the C grid; on hexagons it does not converge.

**The corner condition** For $\mathbf{u}$ constant the inner product must reproduce $\int_C \mathbf{u} \cdot \mathbf{v} = \sum_s s_{Cs} |F_s| \, V_s \, \mathbf{u} \cdot (\mathbf{x}_s - \mathbf{x}_C)$ (exact for fields of constant divergence, $\mathbf{x}_s$ the centroid of the half face $s$). Corner by corner this is $A_{Cv} = R_{Cv}^T N_{Cv}$, $R_{Cv}$ the rows $s|F_s|(\mathbf{x}_s - \mathbf{x}_C)$ of the two half faces (Lipnikov, Shashkov and Yotov 2009, local flux mimetic finite differences), and a symmetric local inner product exists only if every $A_{Cv}$ is symmetric; the scalar weight $\omega_{Cv} I$ needs it isotropic. On level 3 (`corner_A.jl`):

| mesh | ratio of the eigenvalues of $A_{Cv}$ | asymmetry $\|A - A^T\|/\|A\|$ | defect of $\omega I$ |
|---|---|---|---|
| hexagons | 4.8–9.9 | 0.26 | 0.83 |
| cubed sphere | 2.5–3.9 | 0.17 | |
| icosahedral | 1.67–1.68 | 0.095 | |

Symmetrizing $A_{Cv}$ leaves an $O(1)$ error, the non-symmetric $A_{Cv}$ is MPFA-O without the energy pairing. On triangles the weaker consistency of the quadrature (exact for $P_1$ on affine elements) suffices for the second order of the pressure; on hexagons no corner rule of this kind is consistent.

**Spurious slow modes** The half faces double the velocity unknowns: the mode with opposite signs on the two halves of a face carries no flux. Eigenvalues of the linear rotating system (depth 10 km, level 2): hexagons 162 cells, 960 half faces, 312 gravity modes and 810 modes slower than $2\Omega$; icosahedral 320 cells, 960 half faces, 628 gravity modes ($\approx 2N$, no checkerboard as on ICON's triangular C grid) and 652 slow modes, about 330 of them spurious.

**Cell-wise mimetic inner product** With the same half-face unknowns, $M_C = R_C R_C^T/|C| + \gamma \, \mathrm{tr}(M_0)/n_s \, (I - N_C (N_C^T N_C)^{-1} N_C^T)$ per cell (`corner_mfd.jl`, Brezzi, Lipnikov and Shashkov) is consistent on any polygon and symmetric, with the Coriolis force of the cell vectors $R_C^T U/|C|$, but $M$ couples all half faces of a cell and its inverse is global (one sparse SPD solve per stage, well conditioned). Williamson 2 on hexagons 3 and 4: $\gamma = 1$: 0.099, 0.032; $\gamma = 10$: 0.015, 0.0042; $\gamma = 30$: 0.0054, 0.0015. The error falls as the stabilization suppresses the zero-sum modes: the extra unknowns do not help.

**Literature** Egger and Radu: the mass-lumped $\mathrm{BDM}_1 \times P_0$ for acoustic waves (Numer. Math. 145, 2020, arXiv:1803.04238; simplices, first order, $O(h^2)$ pressure by super-convergence, leapfrog), the second-order multipoint flux elements on hybrid meshes (SINUM 58, 2020, arXiv:1812.03938; vertices and barycentre as quadrature points, $\mathrm{RT}_1$, $\mathrm{BDFM}_2$ and enriched spaces with $P_1$ pressure, affine elements only), their use for waves in $H(\mathrm{div})$ (arXiv:1912.07057) and Yee-like schemes for Maxwell (arXiv:2209.10432, 2306.01182). None treats non-affine cells or general polygons; spherical triangles deviate from their chords by $h^2/8a$, a smooth perturbation of affine maps, hexagons are not near-affine images of a reference element.

**The kite elements** The kite model (`docs/paper/kite.tex`) gets its local mass from the spaces, not from inexact quadrature: the velocity is $H(\mathrm{div})$ in the dual cells and broken across the dual faces, so the exact mass matrix is block diagonal per dual cell, and the height mass is diagonal by the Radau nodes. The corner scheme above is essentially its lowest pair $Q_0 \times \mathrm{RT}_0$ with the dual-face unknowns removed and the mass lumped. Two conditions fix the lowest consistent order:

1. *Gradient.* With one height per primal cell ($Q_0$) $B^T\pi$ has no entries on the dual faces, and the block of a dual cell has to produce $\nabla p \cdot \mathbf{n}$ there from the primal half faces alone, the corner condition again; the kite paper measures $7.4, 7.6, 7.7$ for the gradient error of $Q_0 \times \mathrm{RT}_0$ on hexagons. A height continuous and non-constant inside the primal cell ($H_p$, $p \geq 1$) forces the dual-face unknowns through the dual-face term of $G$.

2. *Divergence.* Kites are bilinear images of the square that do not become parallelograms under refinement (on a regular hexagon the angles stay $120°, 90°, 60°, 90°$), and on such quadrilaterals the mapped $\mathrm{RT}_0$ and $\mathrm{BDM}_1$ lose the approximation of the divergence (Arnold, Boffi and Falk 2005); $\mathrm{RT}_1$ keeps first order.

The lowest pair with both operators consistent is therefore $\mathrm{RT}_1 \times Q_1$ (60 velocity and 13 height unknowns per hexagon); $\mathrm{BDM}_1 \times Q_1$ and $Q_1^2 \times Q_1$ of Wess et al. (36 and 13, the latter with $2 \times 2$ blocks) have a non-converging divergence but still converge at second order for the geostrophic state. All these pairs have more than twice as many velocity as height unknowns and so divergence-free fields orthogonal to the curls ($8E - N$ for $\mathrm{RT}_1 \times Q_1$), the counterpart of the spurious slow modes of the corner scheme. One height value per polygon with a local, symmetric velocity mass is consistent only on affine cells; on hexagons the choice is between the two-point gradient (local, energy pairing, consistent on orthogonal meshes), a global mimetic mass matrix, and a richer height space as in the kite elements.

# 10 Tests and examples

| test set (`test/polyhedral.jl`) | checks |
|---|---|
| cell complexes and the integration of monomials | exact moments (cube, prism, non-convex prism), divergence theorem, closure and volume of spherical shells, non-planar faces |
| polynomial reconstruction and transport | polynomials reproduced up to degree 3, tangential basis, mass to round-off, order 4 of the steady transport with degree 3 |
| interaction regions and shallow water | exactness of `G`, `T` for linear fields, cell vectors, Williamson 2 on hexagons and triangles, mass, no allocations, MPI 1 vs. 3 ranks bitwise |
| cut cells | closure, terrain faces on the surface, volume, merging, zero flux through the terrain, free stream, mass, order of the steady transport |
| Euler equations | a stratified atmosphere at rest (not the reference) balanced exactly and kept at rest for a day, surface pressure, mass, the layout of the columns |

Run with `julia +1.12 -project=. test/runtests.jl`. The test files use the packages loaded by `runtests.jl`; to run only these test sets, `julia +1.12 -project=. -e 'using Test, dmplex, LinearAlgebra, SparseArrays, StaticArrays, Random, KernelAbstractions, MPI; include("test/polyhedral.jl")'` (without `MPI` the MPI subtest fails with `MPI` undefined and the later test sets do not run). The MPI subtest starts `mpiexec` with the repository environment, which needs an instantiated `Manifest.toml` for that Julia version (`Pkg.instantiate()`). MPI.jl then uses its own MPICH (`MPICH_jll`,

with its `mpiexec`); a `LocalPreferences.toml` selecting a system MPI is read only when `MPIPreferences` is a dependency of the active project. On the Mac mini all five test sets pass (261 tests, 2.5 minutes).

| `examples/polyhedral/` | |
|---|---|
| `galewsky_poly.jl` | Galewsky jet, MPI and GPU, Figure 3 |
| `mountain.jl` | Euler equations over a hill on cut cells, Figure 5 |
| `figure_data.jl` | data of Figures 1–4 |
| `plot_doc.py`, `plot_mountain.py` | the figures (matplotlib) |
| `../baroclinic_wave.jl poly`, `../plot_surface.py` | baroclinic wave, Figure 6 |
| `corner/` | corner unknowns and the cell-wise mimetic inner product, Sec |